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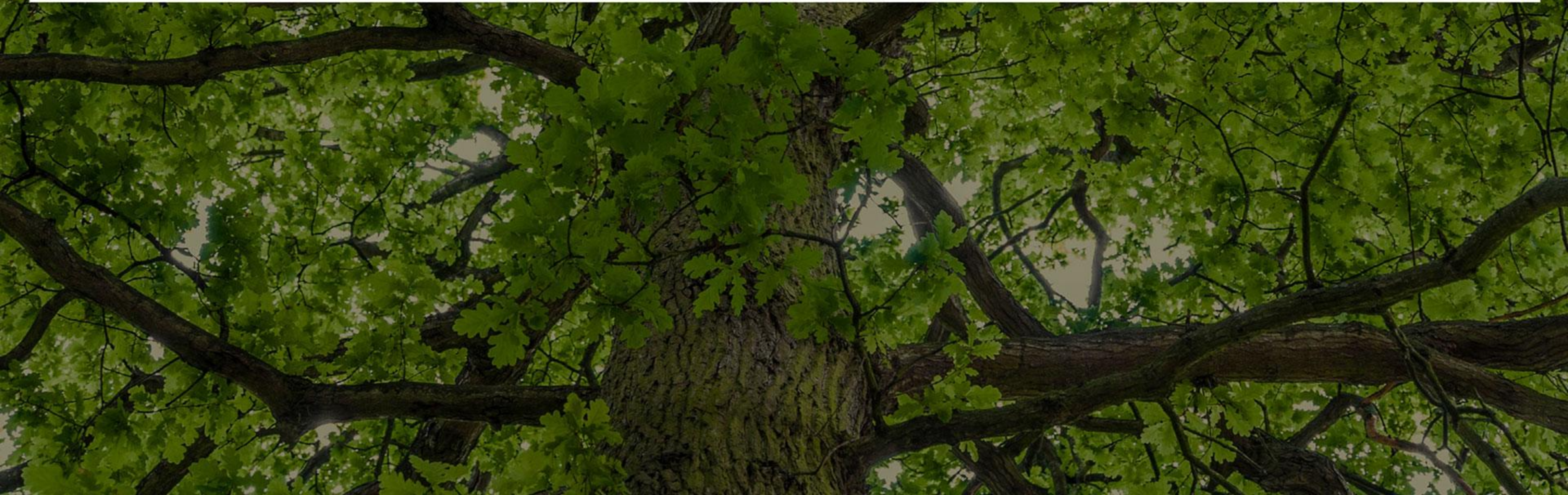
Experimental design concepts

Statistics 244



Observations or experiment?

- **Observational study**
  - A representative sample of units for study is obtained from a specific population
  - Variables of interest are recorded and analysed to reach a conclusion
- **Manipulative, designed experiment (trial):**
  - A representative sample of experimental units is obtained from a specific population
  - One or more ***treatments are deliberately applied*** to the experimental units according to a pre-considered design
  - The responses are recorded and analysed to reach a conclusion

Estimate the relationship between the *independent variables* (treatments) and the *dependent variable* (response)

Goals of experimental design

Simultaneously…

Get a **better answer**:

- More valid
- More accurate
- More precise
- More sensitive
- More useful

Use **fewer resources**:

- Experimental units
- Materials
- Time to get results
- Costs

Minimize ***confounding***, which is when differences due to experimental treatments cannot be separated from other factors that might be causing the observed differences, resulting in possible alternative explanations for the results

https://online.stat.psu.edu/stat200/lesson/1/1.4/1.4.1

Understand components of ***variation*** present, so that they can be eliminated or managed appropriately

Components of total variation

Natural variation	Natural, uncontrollable differences between individuals or units in a population	Cannot be avoided "Even out" effect by allocating units randomly to treatment groups or using suitable experimental designs Estimate using a uniformity trial (pilot test)
Treatment effect	Variation caused by applying different treatments to different groups	What we are looking for; Ensure we can claim the variation was caused by the treatment and not other factors (avoid confounding) through good design and careful execution of the plan
Sampling variation	Samples are subsets of the population and will vary from each other and from the population	Cannot be avoided; means differ between samples from the same population Can be reduced during design by increasing sample size Can be estimated by standard errors of the estimates
Selection bias	Sample does not represent the population, but is "skew" in some way	Avoidable through careful design, proper randomization, etc Limits ability to generalize and apply results Effect of bias cannot be estimated
Observation errors or variation	Instrument errors Human error and/or variability	Occurs during the data collection, cannot be measured Cannot prevent entirely, but can be minimized by careful execution, training, etc
Measurement "errors"	Natural variation between readings taken using the same instrument	Cannot be avoided, but minimized Can be estimated using repeated readings

Experimental design terminology

Experimental unit:

The specific unit, area (plot), individual or quantity of material to which each treatment combination is applied

Treatment (factor):

A deliberate manipulation of experimental units, e.g. apply fertilizer, vary temperature

Treatment levels (factor levels):

The different quantities or categories of each treatment (factor) that are applied to the experimental units, e.g.

- *Quantities*: if treatment: fertilizer, levels: different concentrations of fertilizer used
- *Categories*: if treatment: yeast used, levels: different strains of yeast

Response:

The *outcome* of interest being measured, e.g. time to germination, yield, mass gain

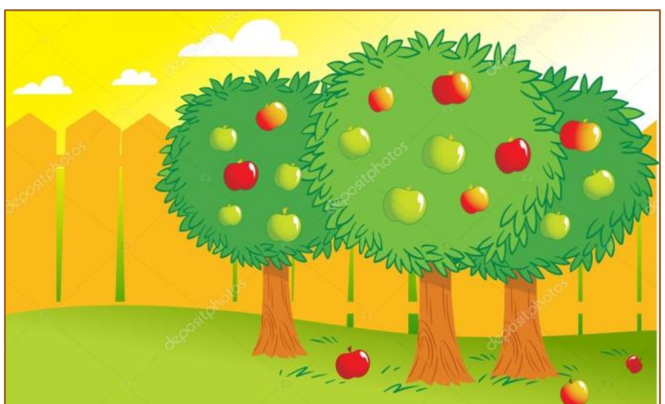
The responses typically studied are *quantitative*; methods exist for other types of response but are not covered here

Replication

What it is: Repeated observations obtained by applying complete treatment combinations to multiple independent experimental units each

Why is it needed?

- Analyses of designed experiments usually compare the variation **between** treatment groups to the *inherent variability* between experimental units **within** each group. Replicate observations are needed to estimate the variability **within** each group
- Increases the precision of the estimates
- Helps to avoid confounding treatment differences with other systematic differences between experimental units



Pseudo-replication occurs when multiple measurements are taken from essentially the same experimental unit, in stead of from different experimental units; it limits the conclusions that can be drawn from the analysis

Suppose types of fertilizer are investigated. A single experimental unit is a tree. This study looks at the quality of apples. Multiple apples from the same tree are examples of ***pseudo-replications***.

Controls

Questions:

- What would have happened anyway if the experimental manipulation had not been done?
- Treatments may be different or not, but are they better/worse than no treatment?

Solution: Include a group in the experiment which does not receive any treatment

Cannot claim the treatment **caused** the effect unless there was a **control group** which received no treatment in the experiment

Randomized control trials:

- Considered the "gold standard" in many fields
- Keep as many factors as possible constant across treatment groups
- Randomly assign units to either treatment group or control group

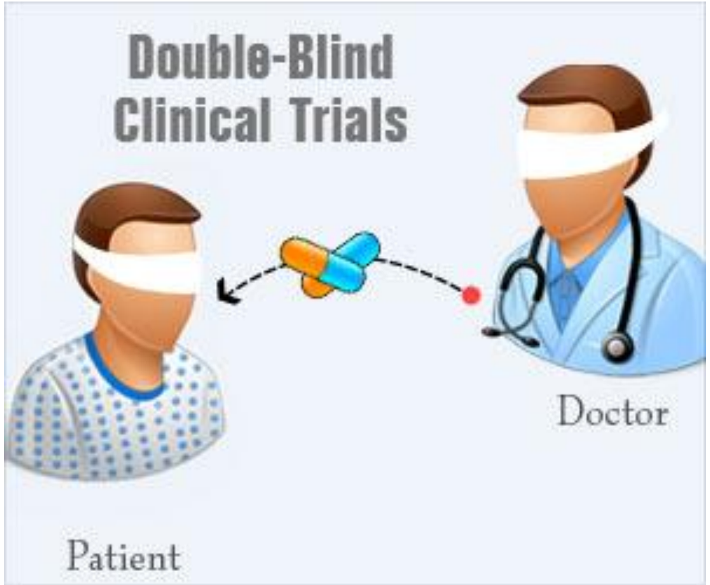
Enter human nature: reducing bias

Single blind

The researcher knows which experimental units are in the various treatment groups.

Double blind

The researcher does **NOT** know which experimental units are in the various treatment groups.



Randomisation

Two kinds of randomisation are important to ensure **independence** between units:

- **Experimental units** within each treatment should be a *representative, random sample* from an appropriate population of experimental units
  - Need to clearly define the population we are taking the random sample from – can only make conclusions about this population
  - Random sample is one in which each individual or unit in the population has an equal chance of being included in the sample
- **Treatments** should be *assigned to experimental units*, or experimental units assigned to treatment groups, randomly
  - No subjective pattern of inclusion, exclusion or how they are treated occurs across experimental units
  - Systematic differences between experimental units that might confound the interpretation of treatment effects are minimized

Uniformity trial

What it is:

- An experiment in which all the experimental units are treated the same
  - Standard treatment or no treatment (control)
- No null hypothesis is under consideration

Sometimes called a blank or dummy experiment, or a pilot study

Why is it needed?

- Supplies an estimate of the natural variance σ^2, which
  - describes natural variability among experimental units
  - can be used to estimate the number of replications needed
- Also helps to test out the logistics and practical aspects of carrying out the experiment

A few examples

The following slides each consist of an example with limited information. From the given information identify the following:

- the treatment,
- number of treatments (k),
- experimental unit,
- number of replications per j^{th} group (n_j),
- total number of experimental units (n),
- and a possible null hypothesis for the mean of the experiment.

A few examples

1. One of two different drugs are given to 7 male rabbits each, and their blood-clotting time is recorded.
   - Treatment
   - Number of treatments (k)
   - Experimental unit
   - Number of replications (n_j)
   - Total number of experimental units (n)
   - Null hypothesis

A few examples

1. One of two different drugs are given to 7 male rabbits each, and their blood-clotting time is recorded.
    - Treatment - drug
    - Number of treatments (k) - 2
    - Experimental unit – male rabbit
    - Number of replications (n_j) - 7
    - Total number of experimental units (n) – 14
    - Null hypothesis – $H_0\colon \mu_1 = \mu_2$

A few examples

2. One of two different fertilizers are given to 10 plants each, and their heights are recorded.
    - Treatment
    - Number of treatments (k)
    - Experimental unit
    - Number of replications (n_j)
    - Total number of experimental units (n)
    - Null hypothesis

A few examples

2. One of two different fertilizers are given to 10 plants each, and their heights are recorded.
   - Treatment - fertilizer
   - Number of treatments (k) - 2
   - Experimental unit - plant
   - Number of replications (n_j) - 10
   - Total number of experimental units (n) – 20
   - Null hypothesis – $H_0\colon \mu_1 = \mu_2$

A few examples

3. 16 cockroach eggs are hatched at three different temperatures (8 eggs per temperature setting), the time for a egg to hatch is recorded.

- Treatment
- Number of treatments (k)
- Experimental unit
- Number of replications (n_j)
- Total number of experimental units (n)
- Null hypothesis

A few examples

3. 16 cockroach eggs are hatched at three different temperatures (8 eggs per temperature setting), the time for a egg to hatch is recorded.

- Treatment – temperature
- Number of treatments (k) - 3
- Experimental unit – cockroach egg
- Number of replications (n_j) - 8
- Total number of experimental units (n) – 24
- Null hypothesis – H_0: $\mu_1 = \mu_2 = \mu_3$

A few examples

4. The number of moths that are caught during the night by four different types of traps are recorded. 40 traps in total were used.

- Treatment
- Number of treatments (k)
- Experimental unit
- Number of replications (n_j)
- Total number of experimental units (n)
- Null hypothesis

A few examples

4. The number of moths that are caught during the night by four different types of traps are recorded. 40 traps in total were used.

- Treatment – type of trap
- Number of treatments (k) - 4
- Experimental unit - trap
- Number of replications (n_j) - 10
- Total number of experimental units (n) – 40
- Null hypothesis – $H_0\colon \mu_1 = \mu_2 = \mu_3 = \mu_4$

A few examples

5. The heights of 7 rugby players and 7 soccer players are recorded and compared.
    - Treatment
    - Number of treatments (k)
    - Experimental unit
    - Number of replications (n_j)
    - Total number of experimental units (n)
    - Null hypothesis

A few examples

5. The heights of 7 rugby players and 7 soccer players are recorded and compared.
   - Treatment - sport
   - Number of treatments (k) - 2
   - Experimental unit – player / athlete
   - Number of replications (n_j) - 7
   - Total number of experimental units (n) – 14
   - Null hypothesis – $H_0\colon \mu_1 = \mu_2$

A few examples

6. Four different rations are given to 10 piglets each, and their increase in mass is measured after three weeks.

    - Treatment
    - Number of treatments (k)
    - Experimental unit
    - Number of replications (n_j)
    - Total number of experimental units (n)
    - Null hypothesis

A few examples

6. Four different rations are given to 10 piglets each, and their increase in mass is measured after three weeks.

- Treatment - ration
- Number of treatments (k) - 4
- Experimental unit - piglet
- Number of replications (n_j) - 10
- Total number of experimental units (n) – 40
- Null hypothesis – $H_0\colon \mu_1 = \mu_2 = \mu_3 = \mu_4$

A few examples

7. Five trees were sprayed with *Remedy A* against pests, five other trees were sprayed with *Remedy B*. Twenty apples are picked from each tree and investigated for symptoms.

- Treatment
- Number of treatments (k)
- Experimental unit
- Number of replications (n_j)
- Total number of experimental units (n)
- Null hypothesis

A few examples

7. Five trees were sprayed with *Remedy A* against pests, five other trees were sprayed with *Remedy B*. Twenty apples are picked from each tree and investigated for symptoms.

- Treatment - spray
- Number of treatments (k) - 2
- Experimental unit - tree
- Number of replications (n_j) - 5
- Total number of experimental units (n) – 10 (note that the apples are considered as pseudo-replicates)
- Null hypothesis – $H_0\colon \mu_1 = \mu_2$